

REVIEW ARTICLE

Hepatitis B Immunization in Adolescents

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This article reviews the epidemiology of hepatitis B in the United States, previous vaccination strategy, and reasons for its failure and issues leading to the recommendation to vaccinate all adolescents. A review of specific hepatitis B virus risk behaviors of adolescents and barriers to vaccinating adolescents is covered. Strategies that favor successful completion of the immunization series are also examined. Hepatitis B infection is an important public health concern for adolescents. The previous vaccine strategy to immunize only individuals thought to be at high risk was unsuccessful, especially because providers of care could not identify these individuals. Furthermore, many individuals thought not to be at high risk for infection were exposed through contacts which could not be identified. Challenges to immunization of adolescents include logistical issues, patient education, cost of the vaccine, and patient compliance. Several of these issues can be addressed by a school-based hepatitis B immunization program. The body of evidence and national policy is rapidly changing to support the recommendation that all adolescents receive the hepatitis B immunization series. The series would be most effective if administered during the middle-school years. A universal adolescent hepatitis B vaccination program would result in the most immediate health benefits and acceleration toward the eradication of hepatitis B in the United States.

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Hepatitis B is one of the most prevalent infectious diseases in the world. Globally, it has been estimated that over 300 million people, more than 5% of the world's population, are chronically infected with hepatitis B (1-3) and over 250,000 people die each year of hepatitis B-associated acute and chronic liver disease (3). The hepatitis B virus (HBV) is a potent carcinogen, and all chronic carriers are at risk of developing hepatocellular carcinoma (4). Chronic HBV infection is estimated to be the etiologic agent for 75-90% of all primary liver cancers worldwide (4); in many highly endemic areas, hepatocellular carcinoma is the most common neoplasm (5). HBV endemic areas include most of Africa, the Pacific Islands, all of Southeast Asia and China, parts of the Middle East, and the Amazon Basin. In these areas, chronic HBV infection usually results from contact with HBV during early childhood through perinatal transmission (6,7) or through close household contact (7,8).

Although the United States is not an HBV endemic area, serologic studies show that as much as 5% of the U.S. population has been infected with the virus (9). Most infections in the United States occur among adolescents and young adults (10) through sexual contact and intravenous drug use (11,12). It is estimated that 300,000 HBV infections occur annually in the United States (13,14), resulting in 20,000 chronic infections and over 4000 deaths (15) due to cirrhosis, acute fulminant hepatitis, or liver cancer (12,13). In the United States, the fatality rate owing to hepatitis B had surpassed all other preventable diseases prior to the implementation of universal immunization programs (400-500 deaths/year from measles and 800-1000 deaths/year from *Hemophilus influenzae* type B infection) (16,17). The economic burden of HBV was estimated to be approximately over \$200 million in 1970 (18).

A safe and effective vaccine to prevent HBV

infection was licensed in the United States in 1982, but despite its availability for over a decade, the incidence of HBV infection in the United States has increased by 37% (19). The paradoxical increase of HBV infections since the introduction of the vaccine is due to the limited scope of the previous vaccination strategy that targeted select high-risk groups including IV drug abusers and persons exposed to HBV infection through sexual or household exposure.

The Immunization Practices Advisory Committee (ACIP) of the Centers for Disease Control and the American Academy of Pediatrics (AAP) revised their guidelines to recommend universal immunization of all infants (19,20). The benefits of infant vaccination, however, will not be seen until the vaccinated cohort reach an age where they might engage in activities that put them at risk for HBV infection. Routine vaccination of adolescents is not currently required, in large part because of the financial (21) and logistical (20) obstacles of such a widespread undertaking. Instead, the ACIP and AAP recommend targeted vaccination for adolescents who display increased risk of infection and universal adolescent immunization in communities with high rates of IV drug use, teenage pregnancy, and sexually transmitted diseases (STDs) (19,20). To reduce the transmission of HBV most rapidly, the AAP stresses the need to vaccinate all children before or during adolescence as soon as resources permit (22).

Universal vaccination of infants alone would have virtually no impact on the disease prevalence for 15 to 20 years, because for most children in the United States, the risk of HBV infection remains low until adolescence (22). Only approximately 8% of acute infections in the United States occur in children younger than 10 years (12). If the current cohort of children and adolescents remains unimmunized, many will continue to be at risk for HBV infection for several decades after the initiation of universal immunization of all infants (22).

This commentary reviews the epidemiology of hepatitis B in the United States, previous vaccination strategy and reasons for its failure, and issues leading to the recommendations to vaccinate all adolescents. Specific HBV risk behaviors of adolescents and obstacles to vaccinating adolescents are covered. This review also outlines strategies that favor successful completion of the three-dose immunization series, including school-based undertakings. Issues concerning indications for vaccination, patient compliance, patient access, and vaccination cost are also addressed.

Background

Hepatitis B is an important public health concern because of its infectiousness, the various routes by which it can be transmitted, and the difficulty in identifying infection through specific clinical symptoms. HBV is present in blood and blood products, saliva (23,24), semen (24), vaginal secretions (25), and wound exudates. The hepatitis B virus is very stable. It can survive up to 1 week in dried blood (22) and is considerably more infectious than HIV. Persons who have close contact with blood and blood products, such as health care workers, hemodialysis patients, and IV drug users, are at risk of infection. HBV transmission through sexual contact is a major route of transmission among the male homosexual population (26–28) and is increasing among the young heterosexual population (11). HBV can be transmitted through close household contact such as sharing razors and toothbrushes (29). Children can transmit the virus to one another through close physical contact such as bites, scratches, or open skin lesions (30).

HBV infection often goes undetected because more than 50% of infections in adolescents and adults are asymptomatic (10,15,31,32). If symptoms do develop, the onset is generally insidious. Many persons experience illness so mild that they do not seek medical attention, or if they do, a diagnosis of hepatitis B is not considered (9,31).

Individuals with acute HBV infection are potentially infectious to others during the course of their infection (<6 months). Chronic carriers are at the heart of the public health concern, because they are potentially infectious to others during their entire lifetime. A variable proportion of infected individuals will become chronic carriers, and the likelihood of developing chronic infection varies inversely with age; between 25 and 50% of children infected before 5 years of age become carriers, whereas 6–10% of acutely infected adults become carriers (13). There are an estimated 1–1.25 million chronic carriers in the United States (19), many of whom may not know they were initially infected, have developed a subsequent carrier state, and are now potentially infectious to others.

Most adults with acute hepatitis B infections recover, with a 1% incidence of fulminant liver disease and death (33). Although no specific treatment exists for acute hepatitis B infection, liver transplants can be performed for patients who have fulminant hepatitis B. Treatment for chronic infection is expensive and is only partially effective (33).

Previous Vaccination Strategy and Reasons for Failure to Reduce Incidence of HBV Infection

The failure of the vaccine to decrease the prevalence of hepatitis B infection is attributed to several factors. The previous immunization strategy for prevention of hepatitis B in the United States was to immunize groups considered to be at high risk for infection (13). Vaccination programs and vaccine usage focused on three risk groups: health care workers exposed to blood, staff and residents of institutions for the developmentally disabled, and staff and patients of hemodialysis units (14). It is estimated that these groups received >85% of the vaccine administered, yet they accounted for only 5 to 10% of acute hepatitis B cases (14). The risk groups that accounted for most of the cases (e.g., IV drug users, homosexual men, and persons acquiring the disease through heterosexual exposure) were not effectively reached (12,14,34). These groups were not vaccinated as a result of a variety of factors, including inaccessibility, noncompliance, lack of vaccination programs, and the inability to identify persons at risk (11,12,35). Furthermore, because over 30% of HBV-infected individuals show no identifiable risk factor for infection (11), they could not even theoretically be included in a target immunization strategy.

Persons exposed to HBV infection through household contact often cannot identify their source of infection or the route of transmission. Most persons with heterosexual exposure cannot identify their source of infection, because infection resulted from exposure to multiple partners or exposure to a contact with subclinical infection (11).

The failure of health care providers to recognize high-risk patients also contributed to the ineffectiveness of the previous HBV immunization strategy (34). Providers are often not aware of groups at high risk for HBV infection and frequently do not identify candidates for vaccination during routine visits (19). A survey conducted in 1985 by the Centers for Disease Control showed that 70% of physicians indicated IV drug abusers as a high-risk group for HBV infection, whereas only 45% identified homosexual men, and very few (10%) identified heterosexuals with multiple partners or heterosexual contacts of carriers (12%) (34). Furthermore, it is unknown whether the medical care providers who are aware of high-risk groups routinely obtain a history detailed enough to identify high-risk behavior (34,36,37).

Epidemiology

The majority of acute hepatitis B cases occur in three groups: IV drug abusers, homosexual men, and persons acquiring disease through heterosexual exposure (38). The rates of HBV infection vary greatly among racial and ethnic groups. Among adolescents and adults in the United States, the incidence of HBV in nonwhite racial/ethnic groups is almost twice that in whites (11). The prevalence of chronic HBV infection is 5–30 times higher among blacks and Asians compared with whites (12). Of the >30% of patients who show no identifiable source for infection, most belong to minority populations and low socioeconomic status (11,39).

The incidence of HBV infection increases rapidly during adolescence, with higher rates among blacks than whites (9). A population-based study showed that blacks start to have an increased prevalence of HBV infection by 12 years of age, whereas whites begin to show an increase at 15 years of age. Within that age range, blacks demonstrate a 3–6-fold higher incidence of HBV infection (9). Racial differences in rates of HBV infection parallel racial differences in rates of sexual activity, which may explain the earlier rise in hepatitis B incidence in black adolescents (9,40).

Several epidemiologic features set adolescents apart from adults. In contrast to adults, for whom the male to female ratio of HBV infection is 2:1, adolescent females outnumber males 1.4:1 (12). Relatively higher rates of HBV infection in adolescent females may indicate their greater physiologic vulnerability than adult females to STDs. A total of 60% of adolescents have no known source for their infections, compared with 30% of infected adults (12). Of the cases of HBV infection in adolescents with a known source of infection, 50% can be attributed to sexual or other person-to-person contact, and 47% to IV drug use (12).

Alter et al. conducted intensive surveillance from 1981 to 1988 of U.S. sentinel counties with rates of HBV infection representative of many areas in the United States (11). During this period, the rate of infection owing to homosexual activity and health care employment declined dramatically (62 and 75%, respectively), whereas that due to heterosexual exposure and IV drug use increased 38 and 80%, respectively (11). Whereas previously, HBV transmission through sexual contact was mainly associated with homosexual activity (26–28), now heterosexual exposure (including sexual contact with

infected persons or with multiple sexual partners) is a more important risk factor (11).

The decrease of HBV infection among health care workers is largely the result of immunization with hepatitis B vaccine and wider implementation of universal blood precautions (11). The decrease among the homosexual male population is attributed to behavioral changes to reduce the risk of acquiring human immunodeficiency virus (HIV) (11,41-48); few individuals in this risk group had actually been vaccinated, because of a lack of programs (12).

Recent studies suggest that many adolescent and young adult homosexual and bisexual men are not as concerned about HIV risks as are older homosexual and bisexual men. Adolescent and young adult homosexual and bisexual men continue to engage in unprotected anal intercourse (49-51), report higher levels of unsafe sexual behavior than do older homosexual and bisexual men (49,52-54), and continue to become infected with HIV (51,55) with infection rates highest among African-American homosexual and bisexual men (51).

Modification of high-risk behavior in the face of AIDS is also less common in heterosexual adolescent and young adult populations (44,56-61), as is risk reduction among urban minorities (44,62). Although some studies indicate a reduction in high-risk sexual behavior by adolescent males (63) and urban minorities (63) in response to concern about contracting HIV, many other studies show that the sexual behavior of adolescents (56,58,59,61-66) and college students (60,67) is not changing dramatically despite increased knowledge about the health risks associated with high-risk sexual practices.

The rate of HBV infection among IV drug users has increased (11), but some data are promising. To reduce the risk for acquiring HIV, IV drug users show an increase in demand for new, unused needles, an increase in needle sterilization (44), and a decrease in needle sharing (44). Although needle exchange programs have shown success (68,69), needle exchange is illegal in 11 states and some cities.

HBV Risk Behavior of Adolescents

The increased risk of acquiring HBV infection is associated with an increased number of sexual partners (70), increased number of years of sexual activity, and history of other sexually transmitted diseases (39,71-74). Heterosexual activity among U.S. adolescents has increased dramatically since the 1970s (63,70,75). In 1970, 4.6% of 15-year-old girls

were sexually experienced, in contrast to 1988, when 25.6% were sexually active (70). The national school-based 1991 Youth Risk Behavior Survey sampled students in all public, private, and parochial schools from grades 9 to 12 to measure health risk behaviors among adolescents (76). Among all students, greater than half had participated in sexual intercourse and almost one-fifth had four or more sexual partners in their lifetime (76,77). Among the students who engaged in sexual intercourse, almost 70% were currently sexually active (within 3 months preceding the survey), and fewer than half had used a condom during their most recent sexual intercourse (76). The prevalence of these behaviors is probably underestimated for the total adolescent population, because adolescents who do not attend school were not sampled, and delinquent youths show marked increases in many HBV risk behaviors (78). The high rates of STDs (79,80) and pregnancy (80,81) in the adolescent population confirm that many adolescents engage in unsafe sexual practices (82). One in eight adolescents is diagnosed with an STD (83). Among sexually active individuals of all ages, 15-19-year-old adolescents have the highest rates of gonorrhea, syphilis, cervicitis, and hospitalizations for pelvic inflammatory disease (PID) (84-86). Adolescents are at higher risk for certain STDs than are persons in other age groups (87) as a result of increased cervical columnar epithelium which is less resistant to chlamydial and gonococcal infection and a relatively unchallenged immune system, because they are less likely than older persons to have been exposed to an STD. Adolescents who have STDs such as herpes, chancroid, and syphilis may be more susceptible to infection because of open genital lesions.

Studies show that as adolescents age, they increase their level of sexual activity and decrease their condom use (88). Use of birth control pills is higher among older than younger adolescents. These findings suggest that although the older group were better protected against unintended pregnancy, they were less protected against acquiring STDs such as hepatitis B.

High-risk individuals may engage in high-risk behaviors. Many injecting drug users share needles and other equipment, trade sex for drugs, and have unprotected sex with numerous partners (89-91). Youth at the highest risk are often homeless and/or runaways, school dropouts, youth offenders, injection drug users, or users of crack cocaine, and therefore may be especially difficult to reach. Alcohol and other noninjection drug use can precede several risk behaviors such as unprotected sex (51,88) larger

number of partners (88), and increased frequency of intercourse (88).

Human immunodeficiency virus and AIDS-reduction research has determined that adolescents are likely to be the most difficult age group to influence toward prevention, largely because of their susceptibility to negative peer pressure, propensity to take risks including sexual and drug experimentation, sense of invulnerability and immortality, and difficulty grasping the long-term adverse consequences of current behavior (92–96). Because the prevalence of hepatitis B infection significantly increases at ages when adolescents begin to experiment with sex and drugs, preventive intervention beyond education and counseling is crucial, and administration of the HBV vaccine is warranted.

Adolescent Vaccination Strategy

Adolescents represent a unique challenge for vaccination. First, the immunization schedule requiring administration of three doses of vaccine over a 6-month period poses certain logistical difficulties. Adolescents are often noncompliant with appointments (97), and successful completion of the immunization scheme may require extra effort to promote compliance. Fitting the three-dose regimen into the academic year favors vaccination provided through school-based clinics or school-wide efforts, although few such extensive undertakings exist (98,99). Thus, the health care provider may have to start the three-dose schedule for many individuals at different times in the academic year. The feasibility for completion of the three-dose regimen (22) around school vacations, examination periods, and so forth should be considered. The 0-1-6-month schedule is recommended for vaccinating adolescents with the age-specific dose of the vaccine (Table 1) (19). The dosage of Engerix-B for adolescents ages 11–19 for the usual dosing schedule of 0-1-6 months previously was 20 μg /dose; the current recommendations are either 10 or 20 μg /dose. Geometric mean titers 1–2 months after completion of the three-dose series (0-1-6 months) are higher for patients receiving the 20 μg dosage, but seroprotection rates are similar for both groups (100).

The AAP recommends that all adolescents be immunized against hepatitis B (22), and the ACIP is currently drafting recommendations for universal adolescent immunization. Pediatric and adolescent health care providers should regard each patient as currently or potentially at risk for HBV infection. For

adolescents, a thorough sexual history should be taken including past sexual experiences. Beside the AAP, others advocate that all sexually active young adults be targeted for vaccination (99). Until the financial resources allow for universal vaccination, at the very least, a target strategy to vaccinate adolescents who demonstrate high-risk behavior is recommended. Until such time as universal immunization is accomplished, the AAP recommends that special efforts be made to vaccinate adolescents who: (a) have sexually transmitted disease(s), (b) have had more than one sexual partner in the previous 6 months, (c) are IV drug users, (d) are sexually active homosexual or bisexual males, (e) are sexual contacts of high-risk individuals, or (f) may have occupational exposure to blood or blood-contaminated body fluids. Immunization should be emphasized for adolescents living in areas where increased rates of IV drug abuse, teenage pregnancy, and/or sexually transmitted diseases occur (20). Requests for oral contraception may be an indication for vaccination, because the use of barrier methods of contraception is low among adolescent users of oral contraceptives (101–103). The health care provider must be skilled in eliciting a personal history detailed enough to identify high-risk behavior, and must convey the importance of receiving the vaccine for those who demonstrate high risk for infection.

Adolescents at highest risk would most likely be least compliant, and asking adolescents to participate in a three-dose immunization series over a 6-month period is likely to result in high dropout rates (20). The health care provider may employ several methods to promote successful completion of the three-dose schedule in this group. Studies show that adolescents are more prone to keep appointments if they feel they are susceptible to infection (104,105), realize appointments are for treatment purposes (106), and understand the negative consequences of not adhering to the regimen (107). The health care provider may discuss the negative consequences of no or incomplete vaccination such as the danger of acute infection; the slim, yet real chance of becoming a chronic carrier; the limited success, high cost, and significant morbidity of the current treatment for chronic carriers; and the likelihood of developing life-threatening liver complications if chronic infection develops.

Adolescents may think they are immune from infection after just one dose of vaccine and may miss or break appointments for subsequent doses. The health care provider should stress that three doses are needed for optimal protection, and it is the third

dose in particular that markedly increases the vaccine's protective capacity (108).

Adolescents will be more compliant with appointment keeping if it is understood that the return appointments are for treatment rather than a non-therapeutic procedure such as screening (106, 109,110). A recent study showed that although pre-vaccination screening may be cost-effective for some high-risk groups, the additional office visit required for prescreening lowers the completion by 22% (111) as well as increases the costs for the entire series.

Clinic-specific factors that lower broken appointments include appointment reminders such as mailed or telephone reminders (112), appointments with the same physician to promote a sense continuity of care and order (97,112,113), and appointments at specific times (112) to limit long waiting times (105,114,115).

Although preexisting health services are not necessary to carry out vaccination (98), the presence of a school clinic may facilitate such programs (98). In a school setting, the vaccine can be coordinated with educational and motivational approaches to encourage student participation (98). Peer counseling services can be used through schools, and is being used in some settings including universities to promote vaccine use and compliance by students. In addition, school-based health personnel can provide follow up for students who do not return consent forms or miss vaccine doses (98). Currently more than 500 school-based clinics are believed to exist. Hepatitis B vaccination programs for adolescents and preadolescents in schools and other settings have been conducted from 1992 to 1994 in California, Louisiana, and Oregon (98). The vaccine was made available on a free and voluntary basis in all instances. Between 82 and 95% of individuals completed the three-dose schedule.

Because the least accessible adolescents may be the most at risk (116), special efforts must be made to reach adolescents not attending school. In Oregon, adolescents at high risk for infection were accessible through juvenile detention centers, school-based primary care clinics, residential facilities for psychosocially dysfunctional children, and family planning and STD clinics (98). To reach more adolescents at risk, the vaccine should also be made available through drug outreach and treatment programs.

Vaccines

Two hepatitis B vaccines, Recombivax HB and Engerix-B, are currently available in the United States

(19). Both are produced by recombinant DNA technology (19) and appear to be equally immunogenic (20). They may be used interchangeably in their respective recommended doses (20) (Table 1). The vaccination schedule most used for adults and children has been three intramuscular injections, the second and third administered 1 and 6 months respectively, after the first (19).

Side effects of the vaccine are rare and, if present, are minimal, consisting primarily of soreness at the injection site. Prevacination serologic testing is generally not cost-effective or indicated except in the cases of high-risk adolescents such as IV drug users and male homosexual or bisexual teenagers (22), or persons from hepatitis B endemic countries, particularly Southeast Asia (111). Immunizing an individual previously exposed to the virus apparently causes no adverse effects (117). However, these individuals should be warned that the vaccine is not protective of previous infection and does not "cure" a chronic carrier. It should be stressed that the vaccine does not make high-risk behavior any safer; they remain at high risk for acquiring all other STDs unless they use appropriate preventive measures (118).

Although the importance of completing the three-dose series to assure (up to 99%) protective immunity should be stressed, partial completion of the vaccination series is not completely ineffective; a

Table 1. Recommended Doses of Currently Licensed Hepatitis B Vaccines (rev. 1995)*

Patient Group	Recombivax HB		Engerix-B	
	Dose: µg	(mL)	Dose: µg	(mL)
Infants of HbsAg-negative mothers and children <11 y	2.5	(0.5) ^a	10	(0.5)
Infants of HbsAg-positive mothers (HBIG [0.5 mL] should also be given)	5.0	(0.5) ^b	10	(0.5)
Children and adolescents 11-19 y	5.0	(0.5) ^b	10	(0.5)
Adults 20 y or older	10.0	(1.0) ^b	20	(1.0)
Dialysis patients and other immunosuppressed adults	40.0	(1.0) ^c	40	(2.0) ^d

^a Pediatric formulation.

^b Adult formulation.

^c Special formulation for dialysis patients.

^d Two 1.0 mL doses administered at one site in a 4-dose schedule at 0,1,2 and 6-12 months.

* Committee on Infectious Diseases, American Academy of Pediatrics, Hepatitis B. In: Peter G, Halsey NA, Marcuse EK, et al, eds., Redbook 1994: Report of the Committee on Infectious Diseases. 1994; 224-238.

study showed that 81% of healthy adolescents achieved protective antibody levels after only two doses of the vaccine (108).

Postvaccination screening for a serologic response is not necessary after routine vaccination of infants, children, or adolescents (19). If a protective antibody response develops after vaccination, vaccine recipients are virtually 100% protected against clinical illness (19). Booster doses of the vaccine are not currently recommended (19); long-term studies show that immunologic memory remains intact for at least 9 years (119) and confers protection against HBV even though antibody titers may fall below detectable levels (120,121).

Cost

The hepatitis B vaccine is the only vaccine for an STD (122) and is considered to be the first effective anticancer vaccine (99,123), because of the strong association between HBV infection and primary hepatocellular carcinoma (4). Yet, cost and unfounded fear of HIV infection have limited its widespread acceptance (122). Fears concerning the earlier plasma-derived vaccine and the transmission of AIDS have been unfounded (40,124) and are nullified, as plasma-derived vaccines are no longer produced in the United States (20). Vaccine prices remain high, however. In the United States, the three-injection course costs \$150–200 for an adult, although the recent lowering of dosage recommendations for adolescents receiving Engerix-B will lower the price of the series. Licensure of new vaccines should decrease the cost (122) of vaccine, and if a universal vaccination program for adolescents were to be implemented, a substantial reduction in the price of the vaccine could result from the high-volume sales it would generate (125). Because cost is a major factor in the cost-effectiveness of any universal hepatitis B immunization program (123), efforts should be made to investigate cooperation between pharmaceutical companies and federal health organizations concerning pricing and volume.

A Model for Future Vaccination Efforts

Vaccinating adolescents and adults is substantially more expensive than vaccinating infants, because of their larger dose requirements and the higher implementation costs of delivering vaccine to target populations (19). Although it is cost-effective, start-up

costs of any universal immunization program will be large (99).

Although vaccinating all children nearing, or in, adolescence should be a priority of the health care provider, this may not be economically feasible at this time. Several models have been proposed most efficiently and effectively to vaccinate the unimmunized cohort when resources allow. The AAP recommends vaccinating children at an earlier age than 11 years (22), because better immune responses are seen in younger persons (126) and younger children require lower doses (22). In studies using three 5 µg doses (half the adult dose) for children under 12 years old, over 99% developed protective antibodies (14). Moreover, because age at acute infection is inversely related to the likelihood of chronic illness, immunizing children as young as possible will have the greatest effect on minimizing the chronic carrier population.

If hepatitis B vaccination is not readily accessible through the adolescent's usual source of health care, then vaccination could be done in schools either by the school nurse or during schoolwide vaccination drives. The first year could involve mass vaccination of all children 10 years of age and older. Universal vaccination of adolescents will result in a more rapid reduction of HBV transmission and infection in this group most at risk of acquiring HBV infection (20). When resources do not permit vaccination of multiple-age cohorts of adolescents, an alternative approach is continuous vaccination of students in a single grade or age cohort (98). Subsequent years efforts would therefore only need to focus on vaccinating each new 10-year-old cohort. Bloom et al. suggest proof of HBV immunization be required for entrance into middle or junior high school (123). Some colleges (99) are contemplating making immunization an admission requirement. Beginning in 1996, the Commonwealth of Massachusetts will require children entering kindergarten to be immunized against hepatitis B.

Unlike most preventive health care programs, a universal infant hepatitis B vaccination program dictates a several-decade perspective (123). Incurring costs of infant vaccination today results in savings in the future by preventing infection in 20 years, and chronic liver disease, hepatoma, and death in 60 years (125). A universal adolescent vaccination program would result in the most immediate health benefits and acceleration toward the eradication of HBV in the United States.

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